

04

Arrhythmias, BLS & Resuscitation (PALS)

Identify pediatric arrhythmias, including common post-cardiac-surgery rhythms, and deliver safe CPR and defibrillation.

4. Arrhythmias, BLS & Resuscitation (PALS)

Learning Objectives

- Distinguish shockable (VF, pulseless VT) from non-shockable (asystole, PEA) arrest rhythms and treat accordingly.
- Recognise sinus tachycardia, SVT, and ventricular tachycardia by rate, regularity and QRS width.
- Prepare and administer emergency arrhythmia drugs at correct pediatric doses.
- Deliver high-quality single- and multi-rescuer CPR and operate a defibrillator/AED safely.
- Recognise arrhythmias specific to the post-cardiac-surgery child (junctional ectopic tachycardia, complete heart block).

1. Arrest Rhythms — Shockable or Not?



Asystole

No pulse · NOT shockable · CPR + epinephrine



PEA

No pulse · NOT shockable · CPR + epinephrine + treat cause



VT / VF (pulseless)

No pulse · SHOCKABLE · CPR + defib + epi + amiodarone

- 1 Assess ABCs
- ↓
- 2 Monitor artefact vs true rhythm?
- ↓
- 3 Call for help / Code Blue
- ↓
- 4 Start CPR if no pulse
- ↓
- 5 Prepare defib / cardioversion
- ↓
- 6 Give drugs per PALS

2. Rhythm Recognition

Rhythm	QRS	Regular?	Pulse?	Initial Rx
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Sinus tachycardia	Narrow	Yes	Yes	Treat cause
SVT	Narrow	Yes	Yes	Vagal → adenosine → cardioversion
VT (with pulse)	Wide	Yes	Yes	Amiodarone / sync cardioversion
VT/VF (no pulse)	Wide	No	No	CPR + defib + epi
Bradycardia	Narrow	Regular	Y/N	Atropine ± CPR

**ANY NEW RHYTHM — BEDSIDE CHECKLIST**

1) Sinus? Every QRS after a normal P. 2) Rate, QRS width, PR, regularity. 3) Pulses/SpO₂/hemodynamics first, whatever the rhythm. 4) Correct acidosis, sugar, temp, electrolytes (Ca/K/Mg). 5) Know your drugs. 6) Defibrillator ready.

3. Emergency Drugs

Drug	Indication	Dose
Epinephrine	Arrest	0.01 mg/kg IV/IO q3–5min (1:10,000)
Adenosine	SVT	0.1 → 0.2 → 0.3 mg/kg rapid push (max 6/12 mg) + saline flush
Amiodarone	VT, VF	5 mg/kg IV over 20–60 min
Atropine	Bradycardia	0.02 mg/kg (min 0.1 mg) IV/IO

**DRAWING UP ADRENALINE 1:10,000**

Supplied 1 mg/mL (1:1000). Dilute 1 mL drug + 9 mL NS → 1 mL = 0.1 mg. Dose 0.01 mg/kg IV/IO = 0.1 mL/kg of diluted solution.

4. Basic Life Support

**Start**

Compressions
within 10 sec

**Push**

Hard & fast, 100–
120/min

**Avoid**

Hyperventilation

**Allow**

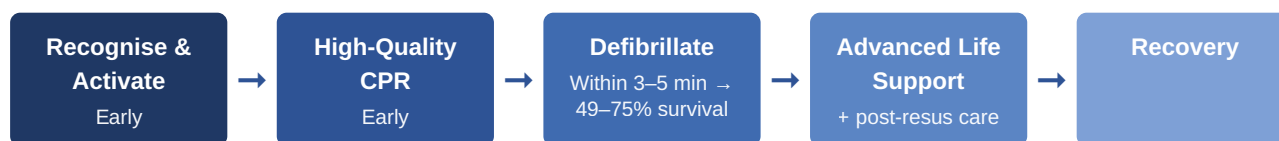
Complete chest
recoil

**Minimise**

Interruptions <10
sec

	Child — single rescuer	Infant — single rescuer
Check	Carotid pulse, scan chest 5–10s	Brachial pulse + resp, ≤10s
Technique	1 hand, lower sternum	2-finger, lower sternum
Ratio (trained)	15:2 ×5 cycles	15:2 ×10 cycles
Ratio (lay)	30:2 ×5 cycles	30:2 ×5 cycles
Depth	≈2 in	1.5 in / ½ AP

Multi-rescuer: R1 compresses, R2 airway + BVM breaths, R3 runs the AED — switch roles whenever the AED analyses.
 Infants: two-thumb encircling technique.



5. Crash Cart & Code Preparedness

Cart at foot of bed; one nurse manages it (drugs, flushes, syringes); document everything. Per-shift check: drawers sealed, drug expiry current, defib/BVM/O₂ functional. After a code: restock immediately, log use.

6. Post-Cardiac-Surgery Arrhythmias



Junctional Ectopic Tachycardia

Narrow-complex, AV dissociation, first 24–72 hr post-repair, poorly tolerated. Cool the patient, optimise sedation, involve cardiology early — avoid pro-automaticity agents.



Complete Heart Block

Can follow surgery near the conduction system. Keep temporary pacing wires & a pacing plan ready.

New post-op tachyarrhythmia: narrow vs wide QRS · check temp/pulse/BP/glucose/electrolytes · 12-lead ECG + repeat echo · inform cardiologist · prep drugs + defibrillator.



LINKS TO THE HANDS-ON LAB

Mega-code simulation, defibrillator/cardioversion practice, and post-cardiac-surgery rhythm-ID case stations — see the Workshop Schedule.

Compiled & adapted from

- PEERS Manual (2025) — Arrest Rhythms, Emergency Drugs, Crash Cart Structure (pp. 44–46); PALS appendix.
- BPCCN Guide — Identifying Rhythm Disturbances for Nurses, case-based ECG stations (pp. 46–52).
- PeC3N — post-operative cardiac arrhythmia case material.

Page-level citations to be added in a subsequent pass. Content has been edited and condensed for print/online study use; refer to the full source manuals for exhaustive detail.